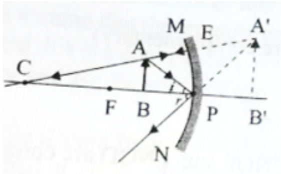
Solution

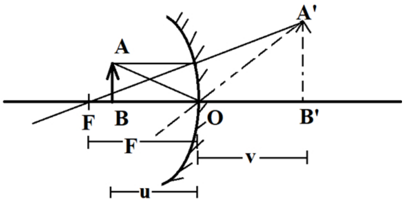
LIGHT -02

Class 10 - Science

1. 

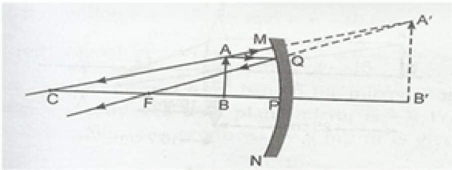
2. u= -12cm
   m= -4
   v= ?
   $\frac{-v}{u} = -4$
   v= 4u
   v= 4(-12)
   v= -48cm
3. Focal length, f = +200cm
   Object distance, u = -100cm
   $\frac{1}{f} = \frac{1}{v} + \frac{1}{V}$
   $\frac{1}{v} = \frac{1}{f} - \frac{1}{V}$
   $\frac{1}{v} = \frac{1}{200} + \frac{1}{100}$
   $\frac{1}{v} = \frac{3}{200}$
   $v = \frac{200}{3} = 66.67cm$
   $m = \frac{-v}{V} = \frac{-66.67}{-100}$
   m = 0.666
   As v is +ve so image is Virtual and is formed behind the mirror.
4. a. Given, R = 5m, f = $\frac{5}{2}$m, u = -20 m and v = ?
      Hence, on applying the values,
      $\frac{2}{R} = \frac{1}{v} + \frac{1}{u}$
      $\frac{2}{5} = \frac{1}{v} - \frac{1}{20}$
      $\frac{1}{v} = \frac{2}{5} + \frac{1}{20}$
      $\frac{1}{v} = \frac{8+1}{20}$
      $\frac{1}{v} = \frac{9}{20}$
      v = $\frac{20}{9}$
      v = 2.8 m
      Nature of the image: Erect, virtual, small and diminished.
   b. Mirror used by dentist is: Concave mirror Reason: It produces enlarged, erect and virtual images of the object, so it is used to see the large image of teeth of patients.
5. *u* = - 10 cm. [u is always negative]
   v = ?; *m* = - 2 [Real image]
   $m = \frac{-v}{u} or -3 = \frac{v}{10}$
   $V = -30\ cm$
   Image will be formed at 30 cm from the mirror on the side of the object.
6. i. The focus of a concave mirror is a point on the principal axis through which the light rays, incident parallel to the principal axis, pass after reflection from the mirror.
   ii. The focus of a convex mirror is a point on the principal axis through which the light rays, incident parallel to the principal axis, appear to pass after reflection from the mirror.
7. The two laws of reflection are:
   i. The incident ray, the reflected ray and the normal at the point of incidence, lie in the same plane.
   ii. The angle of incidence is equal to the angle of reflection (∠i = ∠r)

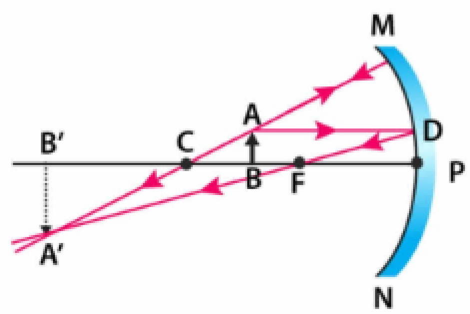
8. Concave mirror is used in a solar furnace. The solar furnace is placed at the focus of the large concave reflector the concave reflector focuses the Sun's heat rays on the furnace and a high temperature is achieved.
9. We have to do calculations for both the concave and convex mirror as the type of mirror is not specified here.
   For concave mirror:
   Focal length, f = -20cm
   Magnification, m = $-\frac{1}{3}$
   Since, magnification, m = $-\frac{v}{u}$
   Magnification, m = $-\frac{1}{3} = -\frac{v}{u}$
   $\Rightarrow$ v $= \frac{u}{3}$
   By mirror formula,
   $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
   $\frac{1}{f} = \frac{3}{u} + \frac{1}{u} = \frac{4}{u} \Rightarrow$ u = 4f
   = 4(-20) = -80cm
   The placement of object should be at 80cm in front of the concave mirror.
   For convex mirror:
   Focal length, f = +20cm
   Magnification, m = $+\frac{1}{3}$
   Since, magnification, m = $-\frac{v}{u}$
   Magnification, m = $\frac{1}{3} = -\frac{v}{u}$
   $\Rightarrow$ v = $-\frac{u}{3}$
   By mirror formula,
   $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
   $\frac{1}{f} = \frac{-3}{u} + \frac{1}{u} = \frac{-2}{u} \Rightarrow -2f$
   = -2(20) = -40cm
   The placement of an object should be at 40cm in front of the concave mirror, to get a diminished, virtual and erect image.
10. f = +15 cm, u = -10 cm.
    1/f = 1/v +1/u
    1/v = 1/15 +1/10
    1/v = 5/30
    v = + 6 cm.
    Magnification = -v/u = 0.6.
    The image is formed 6 cm behind the mirror, it is a virtual and erect image and diminished.
11. He will have to move the screen towards the lens to get a sharp image of the object on it again. Magnification of the image decreases.
12. **Step 1: Given**
    Object distance (u) = -10cm (object distance will be negative, as it is placed on the left side of the mirror)
    Focal length(f)= -15cm (focal length of the concave mirror is taken negatively)
    Step 2: Calculate the magnification and distance of the image,
    (i) Position of the image.-
    Mirror formula $\frac{1}{v} + \frac{1}{u} \ = \ \frac{1}{F}$
    Substituting given values
    The image will be formed 30cm behind the mirror as v is positive.
    (ii) Size of image (m) -
    The magnification formula - $\frac{-v}{u} = \ \frac{-\ 30}{-10} = 3$
    This means that the image is highly magnified.
    (iii) Nature of image - Since v is positive image will be virtual and erect
    Step 3: DIAGRAM



The position of the image is 30cm behind the mirror. Its size is highly magnified, erect, and virtual in nature.

13. Object must be placed in front of concave mirror between its pole and principal focus at a distance less than 15 cm.
The image formed will be virtual and erect. The size of the image is larger the object. The ray diagram is as follows:



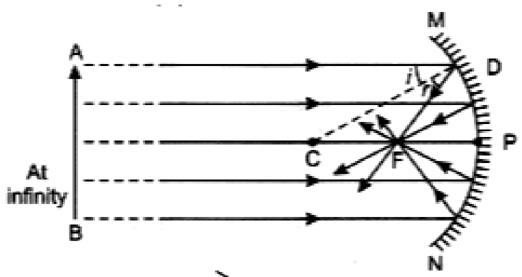
14. 

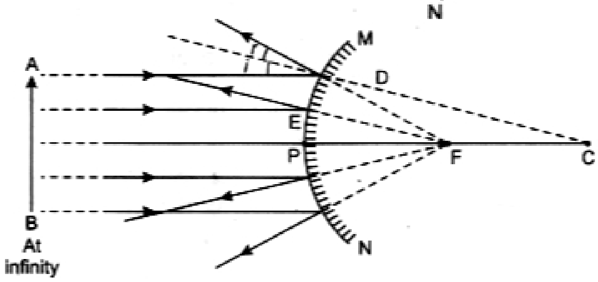
15. Mirror formula is $\frac{1}{f} = \frac{2}{r} = \frac{1}{v} + \frac{1}{u}$
when *f* is the focal length, *r*, the radius of curvature; *u*, the distance of object and *v*, the distance of image from pole of the mirror.
Mirror formula holds good for all types of mirrors i.e. for plane, convex or concave.

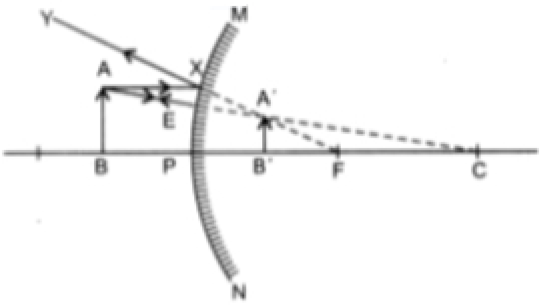
16. i. Real images are formed by the concave mirror. So, he must use a concave mirror.
ii. Given, distance of object, u = -12 cm, distance of image, v = - 48 cm
As, magnification, m $= \frac{-v}{u} = -\frac{(-48/)}{(-12)} = -4$
The negative sign indicates that the image formed is real and inverted.
iii. The image formed on the screen will be 36 cm away form the object.
iv. Real, Inverted and Enlarged image of the candle is formed beyond the Centre of Curvature (C)

17. Image distance, v = - 30 cm
Magnification, m = - 2
Magnification produced by a mirror,
m = $-\frac{v}{u}$
-2 = $-\frac{(30)}{u}$
u = -15 cm
Now, using mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
$\frac{1}{f} = -\frac{1}{30} - \frac{1}{15}$
$\frac{1}{f} = \frac{-1-2}{30}$
$\frac{1}{f} = \frac{-3}{30} = -\frac{1}{10}$
f = -10 cm
If the object is shifted 10 cm towards the mirror then, u = -5 cm, i.e., object is between pole and focus, thus image formed will be virtual, erect and magnified.

18. The expression relating the distance of object 'u', distance of image 'v' and focal length 'f' for a spherical mirror, is called the spherical mirror formula. It is represented as:
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

19. a. 

b. 

20. a. 

b. The three characterstics of the image formed are:

i. Virtual and erect

ii. diminished

iii. formed behind the mirror

c. A convex mirror is used as reflector in street lamps.

21. i. Given, h_o=1cm, h_i= -1.5 cm, u= -15cm

As we know, m = $-\frac{v}{u} = -\frac{h_i}{h_o} \Rightarrow \frac{-v}{-15} = -\frac{1.5}{1}$

$v = -15 \times 1.5 = -22.5cm$. Hence, the imgae will be formed 22.5 cm in front of mirror.

$\therefore \quad m = \frac{-v}{u} = -\left(\frac{-22.5}{-15}\right) = -1.5$

ii. (a) It is used in making shaving mirror.

(b) It is used by dentists to see magnified image of patient's tooth.

22. For convex mirror, we have given, u = - 10 m, R =2.0 m

So, f = $\frac{R}{2}$

= $\frac{2.0m}{2}$

Using the mirror formula,

$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

We get, $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$\frac{1}{1.0} = \frac{11}{10}$

or, v = $\frac{10}{11}$

= 0.9 m

Thus, the car would appear at 0.9 m from the convex mirror. We know that

m = $-\frac{v}{u}$

= $\frac{\frac{-10}{11}}{-10}$

Thus, size of the image of the car will be a fraction of $\frac{1}{11}$ the actual size of the car through the convex mirror.

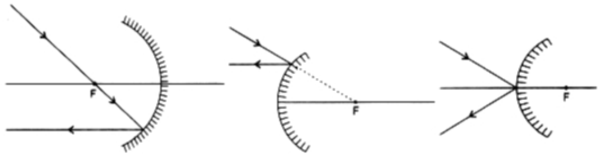
23. Radius of curvature,= R= +32 cm.

Focal length, f = $\frac{R}{2}$

$+\frac{32}{2}cm$

= +16 cm

24. i. The lens is a converging lens.
ii. Such an image is obtained in a magnifying glass and in a lens to correct hypermetropia defect of eye.
iii. The image is virtual, erect and enlarged and is formed behind the concave mirror.

25. 

26. Convex mirrors are used as rear-view mirrors in vehicles to see the traffic at the rear side. Convex mirrors are preferred because
(i) they always give an erect, though diminished image.
(ii) they have a wider field of view as they are curved outwards.
Thus, convex mirrors enables the driver to view much larger area than a plane mirror.

27. The rough focal length of a convex lens is obtained by forming sharp image of a very distant object on a screen. The distance of the screen from the lens gives us the rough focal length of the lens. This method is not applicable to a concave lens, as image formed by a concave lens is virtual and it cannot be taken on the screen.

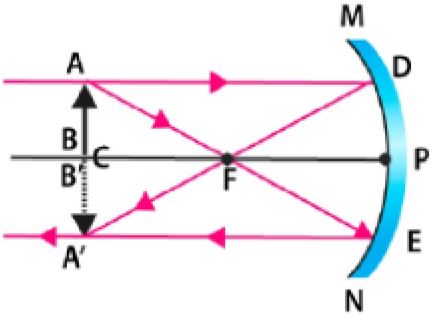
28. $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$
Where f is the focal length of the mirror
U is the object distance
V is the Image distance
U=-25 cm
F=-15 cm
V=?
$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$
$\frac{1}{-15}=\frac{1}{v}+\frac{1}{-25}$
$\frac{1}{v}=\frac{-1}{15}+\frac{1}{25}$
$\frac{1}{v}=\frac{-1}{15}+\frac{1}{25}$
$\frac{1}{v}=\frac{-5+3}{75}$
$\frac{1}{v}=\frac{-2}{75}$
$v=\frac{-75}{2}=-37.5cm$

29. Magnification, m = - 3 (since image is real)
Object distance, u = -10 cm Image distance, v= ?
We know that magnification for the mirror.
Thus, the image is located at a distance of 30 cm in front of the mirror.

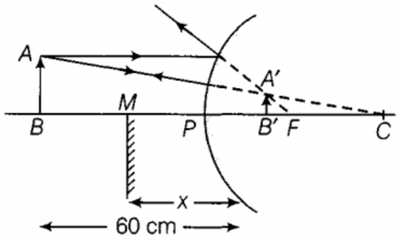
30. Let magnification, $m=-3=\frac{-v}{u}$
Image distance, v = 3u
focal length, f = ?
Object distance, u = -10cm
$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$
v = 3(-10)
v = -30cm.
As value of v is -ve therefore position of image is infront of the mirror by 30cm.

31. i. The mirror is concave.
ii. Given, m = -1, v = -50cm
$\therefore$ Magnification, m = -v/u
$\Rightarrow$ -1 = - (-50)/u
$\Rightarrow$ u = -50cm
So, the distance of image from the object is zero.
iii. Here, the image formed is at the centre of curvature and the focus is half the distance of the centre of curvature, therefore, 2 f = - 50cm $\Rightarrow$ f = -50/2 $\Rightarrow$ f = -25cm
iv. Object distance = image distance = -50cm (minus sign indicates object and image both are in front of mirror)

32. (a) Headlights of a car- concave mirror to give parallel beam of light after reflection from concave mirror.
(b) Side/rear-view mirror of vehicle- convex mirror as it forms virtual erect and diminished image to give wider view field.
(c) Solar furnace- concave mirror to concentrate sunlight to produce heat in solar furnace.

33. 

34. u = -60 cm, f = + 20 cm (convex mirror)



By mirror formula we know :

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{(-60)}$

$= \frac{3+1}{60} = \frac{4}{60}$

v = 15 cm

∴ PB' = 15 cm

For the images in the two mirrors coincide ,let the plane mirror be placed at point M.

As, BM = MB' = MP + PB'

= x + 15...........................(i)

BM = $\frac{1}{2}BB' = \frac{1}{2}(60 + 15)$ = 37.5 ......(ii)

From Eqs. (i) and (ii), we get

⇒ x + 15 = 37.5

⇒ x = 37.5 - 15

= 22.5 cm

35. When the image formed is virtual, erect and enlarged then object should be placed between focus and pole of the concave mirror.

36. Here h_1 = 5 cm; u = - 20 cm, [u is always negative],

r = 30 cm. (convex mirror)

v = ?; h_2 = ?

Using $\frac{2}{r} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{v} = \frac{2}{r} - \frac{1}{u} = \frac{2}{30} - \frac{1}{(-20)} = \frac{1}{15} + \frac{1}{20}$

$\frac{1}{v} = \frac{4 + 3}{60} = \frac{7}{60}$

$v = \frac{60}{7}$

v = 8.56 cm

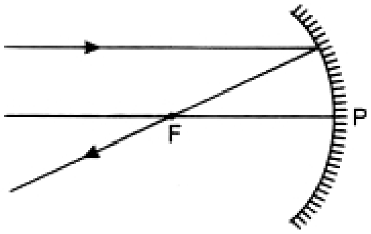
$m = \frac{h_2}{h_1}$ $also\ m = -\frac{v}{u}$

$\frac{h_2}{h_1} = -\frac{v}{u}$ or $\frac{h_2}{5} = \frac{\left[\frac{60}{7}\right]}{-20}$ or $\frac{h_2}{5} = \frac{60}{7\times 20} = \frac{3}{7}$

$h_2 = \frac{15}{7} = 2.14\ cm$

Erect, virtual image 2.14 cm long will be formed behind the mirror at a distance 8.56 cm [between pole and f].

37. A ray of light incident parallel to the principal axis after reflection passed through the principal focus.



A ray of light which passes through the centre of curvature after reflection retraces its path back.

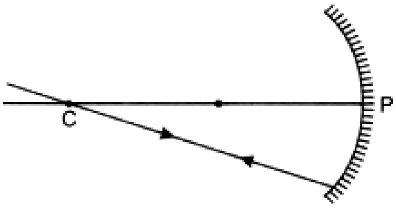
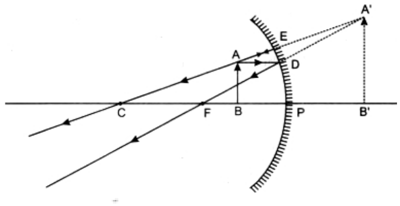
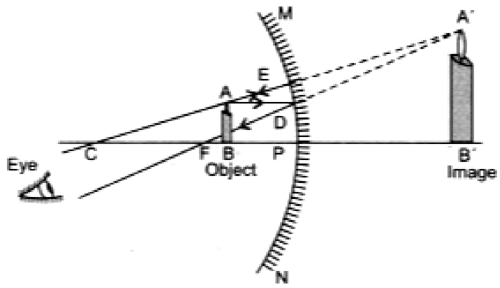


Image of an object placed between pole and focus:



38. a.



b. The image found is :

i. virtual and erect,

ii. magnified and

iii. behind the mirror.

c. A concave mirror in this position can be used as a shaving mirror.

39. The object needs to be moved closer to the spherical mirror to reduce the magnification from $+\frac{1}{2}$ to $+\frac{1}{3}$.

Assume that the image distance is v and the object distance is u. Magnification is calculated as follows: magnification = $-\frac{v}{u}$

Since the initial magnification is $+\frac{1}{2}$, we have: $+\frac{1}{2} = -\frac{v}{20}$ cm

Solving for v, we get: v = -10 cm

Substituting value of v into the magnification equation, we get,

$+\frac{1}{3} = \frac{-10}{u}$

Simplifying, we get: u = 30 cm

Therefore, the object should be placed at a distance of 30 cm from the spherical mirror to reduce the magnification to $+\frac{1}{3}$.

40. In the formula m = $\frac{h_2}{h_1} = \frac{-v}{u}$ the sign of m is determined by the signs of h_1 and h_2. When m is positive, the image is virtual and erect. When m is negative, the image is real and inverted.

41. Radius of curvature, R = + 3.00 m;

Object distance, u = - 5.00 m;

Image distance, v = ?

Height of the image, h' = ?

Focal length, f = $\frac{R}{2} = +\frac{3.00}{2}$

= 1.50 m

Since, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$+\frac{1}{1.50} - \frac{1}{-5.00}$

$\frac{1}{1.50} + \frac{1}{5.00} = \frac{2}{3} + \frac{1}{5}$

$\frac{1}{v} = \frac{13}{15}$

v = $\frac{15}{13}$

= +1.15 m (approximately)

The image is 1.15 m at the back of the mirror.

Magnification = m = $\frac{h'}{h} = -\frac{v}{u}$

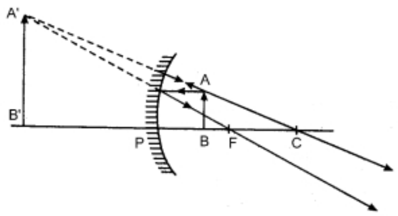
= $-\frac{1.15m}{-5.00m}$

= +0.23

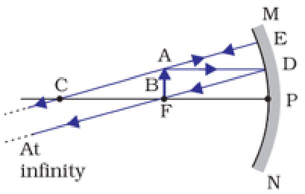
Image is virtual , effect and smaller in size by a factor of 0.23.

42. i. The type of mirror used is convex mirror.
    ii. Convex mirror is used as a side mirror in vehicles. Convex mirrors have a wider field of view as they are curved outwards. Therefore, convex mirrors enable the driver to view much larger area.
    iii. Sunil was a kind and helpful Boy. We learn to help needy people, from Sunil's character

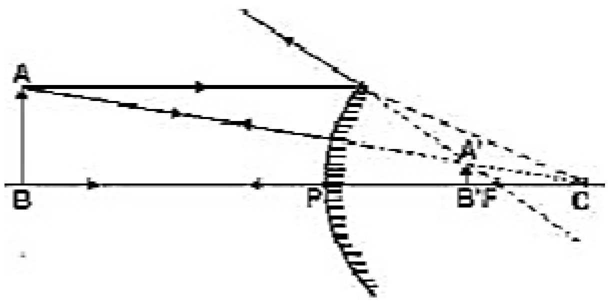
43. Concave mirror produces an erect and enlarged image when the object is placed between pole and focus as shown in the figure.



44. i. Real image, virtual image, magnified image, and image at infinity can be formed in the above diagram.
    ii. Only virtual image can be formed in the above diagram.
    iii. The ray diagram for the image formation by a concave mirror when the object is placed at the focus is as follows:



45. a.



b. The image formed by the convex mirror is always erect, virtual, and smaller in size (diminished).

c. The focal length of a convex mirror is negative. The image distance for a virtual image formed by a convex mirror is positive.

46. a. Concave mirror, as it forms a real image on the same side of the mirror.

b. i. Object distance, u = -15 cm

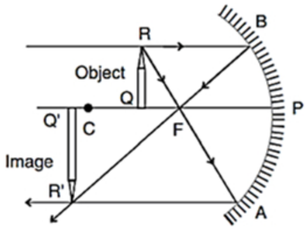
Image distance, v = -60 cm

Magnification, m = $\frac{-v}{u} = \frac{(-60)}{(-15)}$ = -4 cm.

The minus sign in magnification shows that the image formed is real and inverted.

ii. The image is formed at a distance of 45 cm from the object.

c. In this case, the image is formed beyond the centre of curvature. The nature of the image is real, inverted and enlarged.



47. Focal length of a concave mirror, f = -10 cm

Object distance, u = - 15 cm

Object height, = 4 cm

Using the mirror formula, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\frac{1}{v}+\frac{1}{-15}=-\frac{1}{10}$

$\frac{1}{v}=-\frac{1}{10}+\frac{1}{15}$

$\frac{1}{v}=\frac{-3+2}{30}=-\frac{1}{30}$

v = -30 cm

Thus, to obtain a sharp image of the object, the screen should be placed at a distance of 30 cm in front of the mirror.

Now, m = $-\frac{v}{u}=\frac{h_2}{h_1}$

m = $-\left(\frac{-30}{-15}\right)$

$\Rightarrow$ m = -2

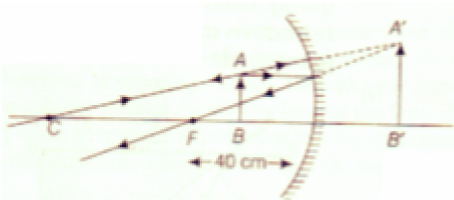
Also, -2 = $\frac{h_2}{4}$

$\Rightarrow$ h_2 = -8 cm

$\therefore$ Height of the image is 8 cm.

48. i. The range should be between 0 - 40 cm, because to have an erect image the object should be between pole(P) of the concave mirror and focus(F).

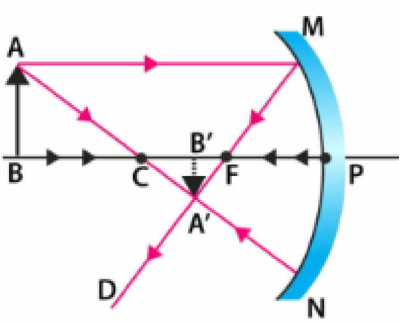
ii. The magnified image will be formed, i.e. The image will be bigger than the object.

iii. 

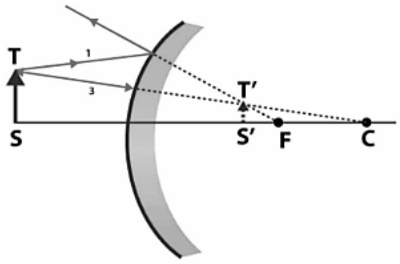
49. Given, magnification produced by a spherical mirror -3, i.e. m=-3.

$\therefore$The four information obtained from this statement are as follows:

i. Real image
ii. Image is inverted
iii. Magnified image and
iv. Mirror is concave

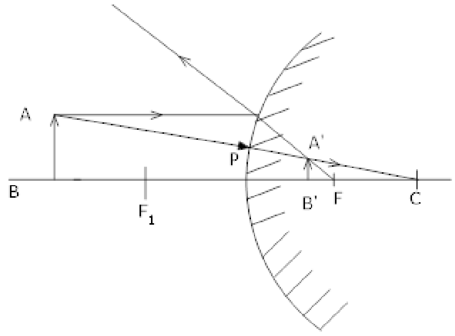
50. 

51. The ray diagram showing the image formation by a convex mirror when an object is placed at the finite distance from the mirror is as follows:



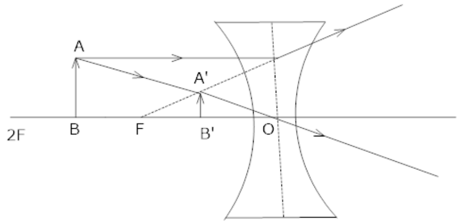
52. Here U = 15 cm f = 10 cm for convex & concave

a.



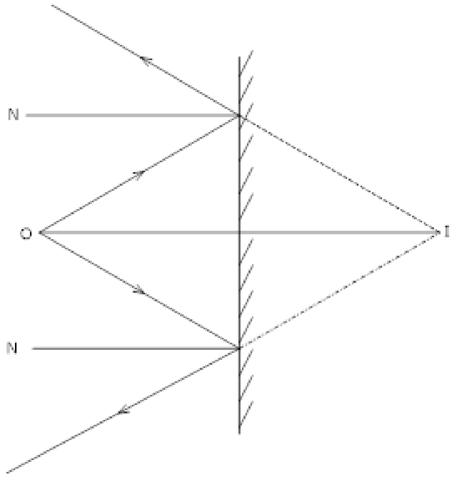
Between Pole and focus virtual & erect Diminished

b.

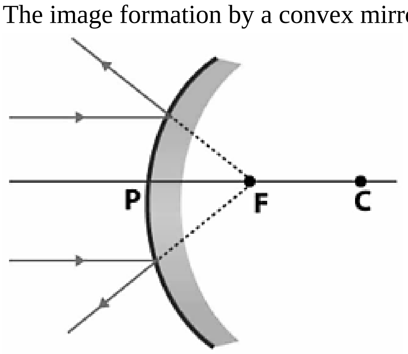


Between optical centre & focus diminished, virtual & erect

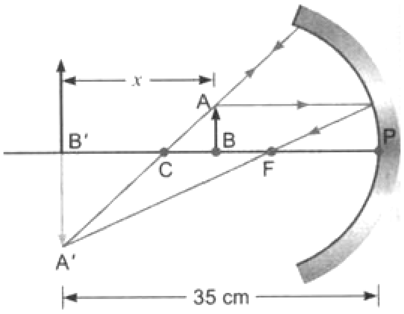
c.



53. The image formation by a convex mirror when an object is placed at infinity is as follows:



54. Here, f =-10 cm; v = - 35 cm (since the image is formed on the wall and distance between the wall and mirror is 35 cm). Image is beyond 2f so object has to be in between F and C.
Let the distance of the object from the wall be x.



Let the distance of the object from the wall be x.

Therefore, u = -(35 - x)

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$

$\frac{1}{-(35-x)}+\frac{1}{35}=\frac{1}{-10}$

or $\frac{1}{35-x}=\frac{1}{10}-\frac{1}{35}$

or $\frac{1}{35-x}=\frac{7-2}{70}$

$\frac{1}{35-x}=\frac{5}{70}$

35-x = 14

x = 49 cm

Therefore, the distance of the object from the wall = 49 cm

55. h_1 = 7 cm, u = - 27 cm, v = ?, f = - 18 cm. (concave mirror)

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u},\ we\ have \frac{1}{v}=\frac{1}{f}-\frac{1}{u}$$

$$or\ \frac{1}{v}=\frac{1}{(-18)}-\frac{1}{(-27)}=-\frac{1}{18}+\frac{1}{27}=\frac{-3+2}{54}=-\frac{1}{54}$$

Now

$$m=-\frac{v}{u}=\frac{h_2}{h_1} or\ \frac{(-54)}{(-27)}=\frac{h_2}{7}$$

$$-2=\frac{h_2}{7}\ or\ h_2=-14cm$$

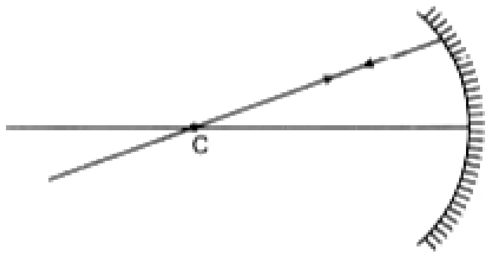
Negative sign of h_2 indicates that image is on the same side as that of the object. It is real, inverted and 14 cm in size.

56. i. The mirror used by the dentist is concave mirror.
ii. The phenomenon of light by which doctor is able to examine Kapil is reflection of light.
iii. The doctor gave the correct advise to Kapil on how to keep his mouth clean and gums healthy. The doctor’s son and doctor was helping in nature. Kapil followed the advise given by the doctor. So, he is an obedient boy.

57. It is a convex mirror since the magnification is positive as well as less than one. Image is diminished and erect.

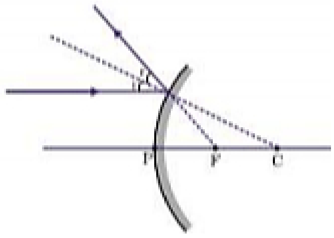
58. It is a concave mirror having focal length – 15 cm and radius of curvature – 30 cm.

59. This is because the angle of incidence is 0^o. That is the ray passing through the centre of curvature is incident normally to the mirror. The angle of reflection should also be 0^o.

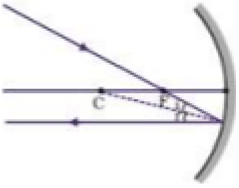


60. The convex mirror is used as a side/rear-view mirror of a vehicle because
i. It forms an erect, virtual, and diminished image of an object placed anywhere in front of it.
ii. It has a wider field of view than a plane mirror of the same size.

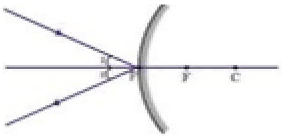
61. i. it is parallel to the principal axis and falling on a convex mirror

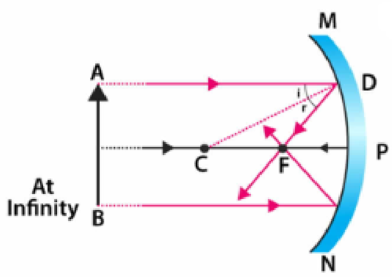


ii. it is falling on a concave mirror while passing through its principal focus



iii. it is coming oblique to the principal axis and falling on the pole of a convex mirror.



62. 

63. Given, Height of the object = 6 cm

Focal length, f = - 30 cm

Object distance, u = - 45 cm

Image distance, v = ?

Height of image, h_i = ?

We have, $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{-30} = \frac{1}{v} + \frac{1}{-45}$

$\frac{1}{-30} + \frac{1}{45} = \frac{1}{v}$

$\frac{1}{v} = \frac{-3+2}{90}$

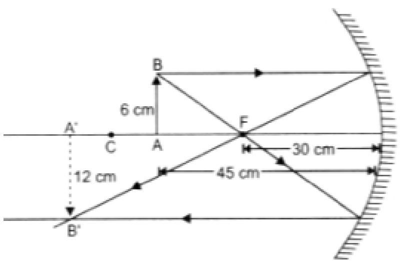
v = - 90 cm

Also, we have m = $\frac{h_i}{h_0} = \frac{-v}{u}$

$\frac{h_i}{6} = \frac{-(-90)}{-45}$

$\frac{h_i}{6}$ = - 2

h_i = -12



64. Concave mirrors are convergent mirrors. That is why they are used to construct solar furnaces. Concave mirrors converge the light incident on them at a single point known as principal focus. Hence, they can be used to produce a large amount of heat at that point.

65. i. h = 3 cm

f = -12 cm

u = -18 cm

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{-12} = \frac{1}{v} + \frac{1}{-18}$

$\frac{-1}{36} = \frac{1}{v}$

image distance = 36 cm

ii. m = $-\frac{v}{u}$

m = $\frac{-36}{-18}$

m = -2

m = $\frac{hi}{ho}$

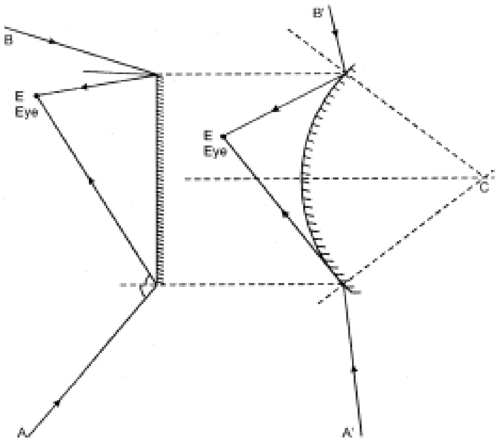
-2 = $\frac{hi}{-3}$

h_i = -6 cm

height of the image is 6 cm.

66. Convex mirrors are used as rear-view mirror in vehicles because of the following two reasons:
    i. 1. A convex mirror always forms an erect, virtual and diminished image of an object placed anywhere in front of it.
    ii. 2. A convex mirror has a wider field of view than a plane mirror of the same size. Thus, convex mirrors enable the driver to view much larger traffic behind him than would be possible with a plane mirror.

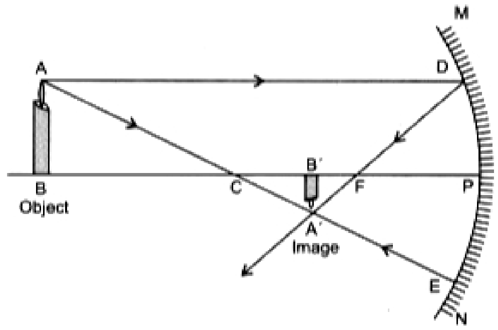
The disadvantage of a convex mirror is that it does not give the correct distance and hence, the speed of the vehicle approaching from behind cannot be estimated precisely.
For the same size of the mirrors (plane and convex), whereas the field of view of plane mirror is from A to B, the field of view of rare traffic is much larger i.e. from A¢ to B¢.



a. Small field of view
b. Large field of view

67. u = –10 cm [u is always negative]; f = 15 cm [convex mirror] v = ?
Using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$, we have $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-10} = \frac{1}{15} + \frac{1}{10} = \frac{2+3}{30} = \frac{5}{30} = \frac{1}{6}$
So v = 6 cm behind the mirror or towards left of the mirror. Image is virtual and erect.

68. u = - 27 cm, f = - 18 cm. h_o= 7.0 cm
1/v = 1/f - 1/u
1/v = -1/18 + 1/27 = -1/54
v = - 54 cm
Screen must be placed at a distance of 54 cm from the mirror in front of it.
h_i/h_0= v/u
$h_i/7$ = +54/-27
h_i = -2 × 7 = -14 cm.
Thus, the image is of 14 cm length and is inverted image.

69. a. 

b. Characteristics of image formed:

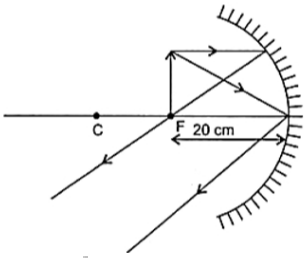
i. Real and inverted.
ii. Smaller in size or diminished.
iii. Formed between the focus F and the centre of curvature C of the mirror.

70. Convex mirror is preferred as rearview mirror in vehicles because it always forms virtual erect and diminished image. If also covers the wider field of view.

71. **Object position:** At C (centre of curvature)

Object distance: 40 cm. If the object is moved 20 cm towards the mirror then,

Position of the image: At infinity, because the focal length of the mirror is 20 cm. If the object is moved 20 cm towards the mirror then its new position would be at the focus of the mirror.



72.